

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of (MSc) Biotechnology Examination March 2019

Course code & Name: MS 713 Cell biology

Date: 18/03/2019 Day: Monday Time: 10:00a.m To 1:00 p.m. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

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- Which of the following cellular processes does NOT involve plasma membrane?
(i) Receptor mediated endocytosis (ii) Vesicular trafficking between ER to Golgi
(iii) Exocytosis (iv) None of the above
- Which of the following technique is used to study lateral mobility of lipids and proteins in bilayer?
(i) Fluid mosaic method (ii) Patch Clamp technique
(iii) FRAP (iv) Electron Spin Resonance
- Which of the following correctly matches an organelle with its function?
(i) ER . . . O-linked glycosylation (ii) nucleus . . . protein trafficking
(iii) Golgi apparatus . . . protein sorting (iv) Lysosome . . . cytoplasmic streaming
- Nucleolus is NOT involved in _____.
(i) formation of nuclear envelope (ii) assembly of ribosomal units
(iii) rRNA synthesis (iv) any of the above

5. Glycophrin is a/an _____.
- (i) ion channel
 - (ii) transmembrane protein
 - (iii) peripheral protein
 - (iv) carrier protein
6. Which of the following is NOT TRUE for Nuclear Localization Signals?
- (i) They direct nuclear import of proteins
 - (ii) They can be localized anywhere across the protein to be transported in nucleus
 - (iii) They do not get cleaved after nuclear import
 - (iv) They are rich in negatively charged aminoacids
7. Role of dolichol in N-linked glycosylation is _____.
- (i) activating sugar residues for glycan synthesis
 - (ii) acting as carrier protein for assembly of glycan
 - (iii) acting as carrier lipid for assembly of glycan
 - (iv) acting as oligosaccharide transferase
8. Which of the following features is/are common to protein translocation in mitochondria and chloroplasts?
- (i) Protein translocation is post-translational
 - (ii) Protein translocation is energy dependent in both organelles
 - (iii) Chaperones are involved in protein transport
 - (iv) All of the above
9. Which of the following ECM proteins have selected prolines and lysines hydroxylated post-translationally?
- (i) Fibronectin
 - (ii) Integrins
 - (iii) Clathrin
 - (iv) Collagen
10. Addition of mannose 6-phosphate on newly synthesized proteins is important for_____.
- (i) Synthesis of glycoproteins
 - (ii) sorting of lysosomal hydrolases
 - (iii) phagocytosis
 - (iv) O-linked glycosylation of proteins
11. Treadmilling is a phenomenon predominant in _____.
- (i) Intermediate filaments
 - (ii) ECM
 - (iii) Actin filaments
 - (iv) Microtubules
12. Intercellular exchange of nutrients is regulated by _____.
- (i) Tight junctions
 - (ii) Septate junctions
 - (iii) Gap junctions
 - (iv) Anchoring junctions
13. Clathrin coated vesicles are involved in _____.
- (i) Protein import
 - (ii) Protein trafficking from trans-Golgi
 - (iii) receptor mediated endocytosis
 - (iv) Both (ii) and (iii)

14. Which of the following is NOT a cytoskeletal protein?
- (i) Occludin
 - (ii) α -actinin
 - (iii) Fimbrin
 - (iv) Filamin
15. Nitric oxide gas signals by binding directly to _____ in the target cell.
- (i) Adenylate cyclase
 - (ii) NO synthase
 - (iii) Guanylyl cyclase
 - (iv) None of the above
16. Insulin signals via _____.
- (i) G protein-coupled receptors
 - (ii) paracrine signaling
 - (iii) receptor tyrosine kinase
 - (iv) activation of Na^+ channels
17. Which of the following is NOT a typical feature of general cell signaling cascade?
- (i) Involvement of secondary messengers in signal transduction
 - (ii) Amplification of signals
 - (iii) Desensitization
 - (iv) None of the above
18. The commitment step near the end of G1 phase of yeast cell cycle is called _____.
- (i) G_1 checkpoint
 - (ii) G_0 checkpoint
 - (iii) Start point
 - (iv) Restriction point
19. Which of the following is a cyclin-Cdk complex?
- (i) APC
 - (ii) MPF
 - (iii) Both of the above
 - (iv) None of the above
20. Which of the following is NOT involved in intrinsic apoptotic pathway?
- (i) Mitochondria
 - (ii) Caspase
 - (iii) FAS ligand
 - (iv) Apoptosome

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Instructions:

- 1. Section I and II must be attempted in TWO ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I

Q - II Answer the following questions as directed 20

- (A) Explain in brief the functions of cholesterol in plasma membrane. (2)
- (B) Answer ANY TWO of the following. (8)
- (i) Discuss various types of membrane transport mechanisms
 - (ii) Describe any one technique to study membrane proteins and mention its application.
 - (iii) Diagrammatically depict the structure of a typical neuronal cell and explain in detail the mechanism of action potential.
- (C) Briefly explain cell cycle checkpoints. (2)
- (D) Answer ANY TWO of the following. (8)
- (i) Discuss the intrinsic and extrinsic pathways of apoptosis
 - (ii) Describe the role of MPF and APC in cell cycle regulation
 - (iii) Describe an experiment to depict the use of yeast as model to understand cell cycle regulation.

SECTION – II

Q-III Answer the following questions as directed 30

- (A) Explain the following (Any SIX) 18
- (i) Assembly of actin filaments
 - (ii) Types of plastids and their significance
 - (iii) Mechanism of G-protein activation
 - (iv) Mechanism of protein translocation in chloroplast
 - (v) Misfolded protein response in ER
 - (vi) SNARE hypothesis

- (vii) Function and synthesis of collagen
- (viii) Tight junctions and Gap junctions

(B) Answer the following (Any THREE)

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- (i) Discuss various types of cell surface receptors.
- (ii) Describe the mechanism of N-linked glycosylation in ER.
- (iii) Discuss the mechanisms of receptor mediated endocytosis.
- (iv) Discuss in detail the composition of extracellular matrix.
- (v) Discuss various types of cell signaling mechanisms.